Chapter 40

Campbell Biology, AP Edition, 12e

Chapter 40

Basic Principles of Animal Form and Function

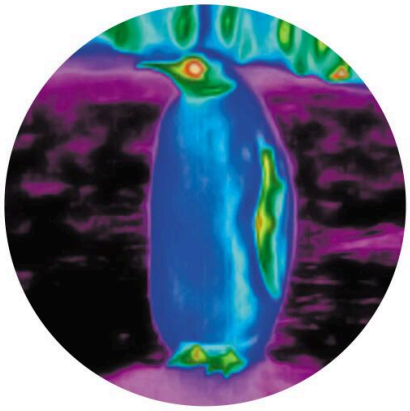
Figure 40.1

Emperor penguins (*Aptenodytes forsteri*) live in Antarctica, Earth's coldest and windiest continent. In summer, these birds catch fish by diving down 500 meters in water only 2°C above freezing. In winter, the females forage and the males incubate eggs while temperatures drop to −40°C and winds gust to 200 km/hr.

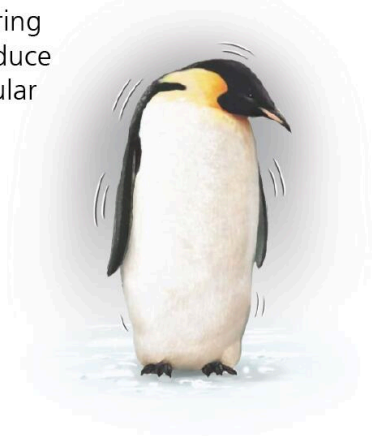
How do animals regulate their internal state even in changing or harsh environments?

Adaptations in **form, function, and behavior** help maintain an animal's internal environment. Adaptations that limit variation in temperature and other internal variables are widespread and diverse. Consider, for example, three adaptations that help an Emperor penguin stay warm:

Form (anatomy): An insulating layer of fat (blubber) reduces heat loss from most of the penguin's body (blue body areas in this thermal image).



Function (physiology): Rapid cycles of muscle contraction and relaxation during shivering produce heat at a cellular level.



Behavior: By packing together in groups of up to several thousand, Emperor penguins greatly reduce their exposure to wind and cold.

Adaptation in form, function, and behavior help maintain an animal's internal environment. Consider, for example, three adaptations that help an Emperor penguin stay warm. Adaptations that limit variation in temperature and other internal variables, such as the solute composition and p H of circulatory fluids, are widespread among animal species. Form (anatomy): An insulating layer of fat (blubber) reduces heat loss from most of the penguin's body. Function (physiology): Rapid cycles of muscle contraction and relaxation during shivering produce heat at a cellular level. Behavior: By packing together in groups of up to several thousand, Emperor penguins greatly reduce their exposure to wind and cold.

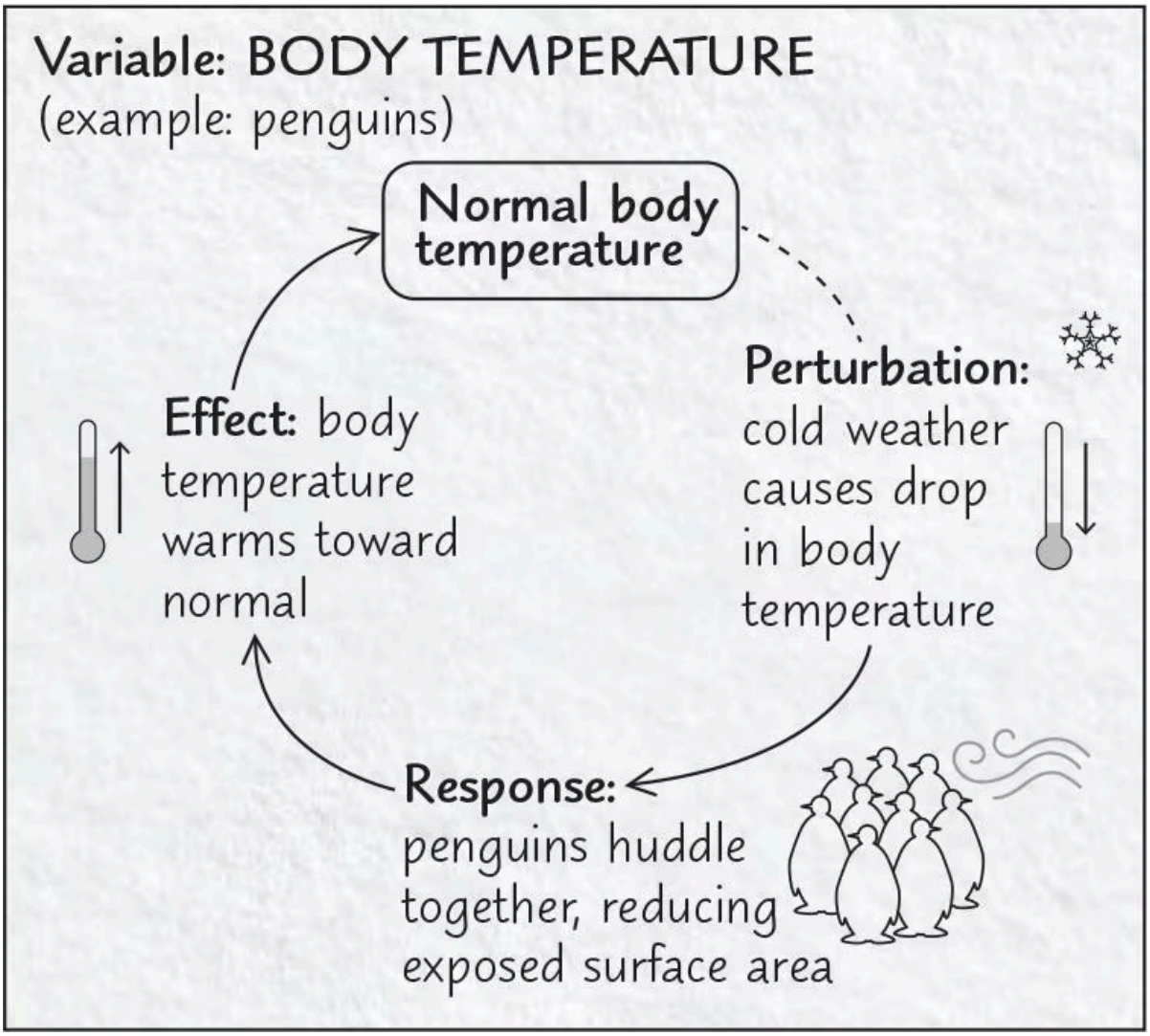
Key Concepts

- **40.1Animal form and function are correlated at all levels of organization****
- **40.2Feedback control maintains the internal environment in many animals****
- **40.3Homeostatic processes for thermoregulation involve form, function, and behavior****
- **40.4Energy requirements are related to animal size, activity, and environment****

All animals require energy and matter to grow, reproduce, and maintain dynamic homeostasis **(BIG IDEA 2)**. They have diverse structures and mechanisms that ensure survival **(BIG IDEA 4)** and that help to define their relatedness **(BIG IDEA 1)**.

Study Tip

Draw a diagram: When you encounter an example in the chapter of how an animal maintains a steady internal state, draw a simple circuit diagram (see example—illustrations are optional!). Label the variable being controlled, a perturbation that affects the variable, the response, and its effect in restoring the normal state.



Get Ready for This Chapter

In this chapter, you will learn about the close fit between anatomy and physiology that arises over the course of evolution. Before beginning this chapter, you should be able to do the following:

- Describe the relationship between surface area and volume (see Concept 6.2, Figure 6.7).
- Describe how adaptation occurs through natural selection (see Concept 1.2, section entitled "Charles Darwin and the Theory of Natural Selection" and Concept 22.2).
- Explain what is meant by an emergent property (see Concept 1.1, section entitled "Emergent Properties").

The following questions provide a quick check of your basic knowledge in these areas.

1. Imagine you have two sets of 8 blocks, each measuring 1 cm along each edge. You arrange one set into a large cube, 2 blocks * 2 blocks * 2 blocks. You arrange the other set into a straight beam, 1 block * 8 blocks. Compared with the beam, the large cube has

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a smaller volume and surface area.
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That's incorrect. Try again.

Incorrect. Review the following material: Each set of blocks has the same total volume 8cm3 but they differ in surface area.

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the same volume and surface area.
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That's incorrect. Try again.

Incorrect. Review the following material: Each set of blocks has the same total volume (8cm3), but they differ in surface area.

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a greater volume and surface area.
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That's incorrect. Try again.

Incorrect. Review the following material: Each set of blocks has the same total volume 8cm3 but they differ in surface area.

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the same volume but a smaller surface area.
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That's correct.

Correct! If you had trouble with this question, review the following material: Each set of blocks has the same total volume (8cm3), but they differ in surface area. The large cube has 24 block sides exposed and thus a surface area of 24cm2, whereas the beam has 34 sides exposed and thus 34cm2 of surface area. Said another way, the large cube has a lower ratio of surface area to volume than the beam (3.0 compared with 4.25).

2. Natural selection refers to

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a mechanism of evolution.
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That's correct.

Correct! If you had trouble with this question, review the following material: Evolution is the change in the genetic make-up of a population over many generations. Natural selection is a mechanism of evolution in which individuals with certain heritable traits tend to survive and reproduce at higher rates than other less well suited individuals.

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a goal of evolution.
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That's incorrect. Try again.

Incorrect. Review the following material: Natural selection is the process by which individuals with heritable traits that make them better suited to their environments survive and reproduce at higher rates than other less well suited individuals in the population.

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the evolution of an individual organism.
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That's incorrect. Try again.

Incorrect. Review the following material: Natural selection is the process by which individuals with heritable traits that make them better suited to their environments survive and reproduce at higher rates than other less well suited individuals in the population.

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the outcome of evolution.
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That's incorrect. Try again.

Incorrect. Review the following material: Natural selection is the process by which individuals with heritable traits that make them better suited to their environments survive and reproduce at higher rates than other less well suited individuals in the population.

3. An emergent property belongs to

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the whole of something but not its parts.
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That's correct.

Correct! If you had trouble with this question, review the following material: Emergent refers to properties that exist in the whole of something, but not in its parts. Thus, for example, a bicycle can be used for transportation, but its parts, such as the pedals, wheels, gears, or frame, cannot. Transportation is thus an emergent property of the assembled bicycle.

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the parts of something but not the whole.
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That's incorrect. Try again.

Incorrect. Review the following material: An example is that transportation is an emergent property of an assembled bicycle.****

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both the parts and the whole of something.
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That's incorrect. Try again.

Incorrect. Review the following material: An example is that transportation is an emergent property of an assembled bicycle.

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the newest part of something.
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That's incorrect. Try again.

Incorrect. Review the following material: An example is that transportation is an emergent property of an assembled bicycle.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 40.1Identify the levels of organization within the animal body, and use examples to show how structure and function are related at each level.
- 40.2Distinguish how negative feedback and positive feedback mechanisms regulate internal conditions.
- 40.3Use examples to demonstrate how endotherms and ectotherms regulate body temperature.
- 40.4Relate the energy requirements and metabolic rate of animals to their size, activity, and environment.